A playbook for a first Rust component in a C++ control plane

Each step carries the measurement that justifies it, and the band below carries what the migration does not buy

1 Choose by interface

Not by size or by how hot it is. What decides the cost is how narrow and scalar the API already is.

MEASURED
11 entry points and 231 lines, already scalar.

2 Design the C surface

An opaque handle, scalar-typed operations, and a status convention. This is the step needing real design decisions.

WHERE THE EFFORT IS
Rich objects, templates, callbacks: design out.

3 Isolate the ABI

Every raw-pointer operation goes in a separate seam crate depending on an unmodified core, under one documented contract.

MEASURED
0 unsafe in the core, 9 functions in the seam.

4 Measure twice

The round-trip, which says what a deployment feels, and the component, which says what the language does. Neither substitutes.

WHY
A round-trip is mostly the OS, not the language.

What this buys

The migrated component stops contributing new use-after-free, double-free, iterator-invalidation, and data-race defects, at compile time, on every ordinary build, and for a run-time cost the measurement bounds at a small fraction of one operation.

What it does not buy: the part most likely to be dropped when the result is summarised

Protection from the host
One address space: a C++ memory error can corrupt Rust-owned data however checked the Rust side was.

A smaller concurrency review
Lock-ordering discipline and review for composed-but-not-atomic operations stay just as necessary.

A free first component
The dominant cost is human. Treat the first as a training investment and instrument it.

Sequence the migration so that components sharing the most mutable state cross over together rather than one at a time.